

Samuel King

PHD STUDENT · GEORGETOWN UNIVERSITY

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Professional Summary

Fifth-year PhD candidate in theoretical computer science with research in algorithms, data structures, and complexity theory. Combines strong theoretical foundations with hands-on experience developing pipelines and predictive models for novel problems in applied settings. Seeking opportunities to apply rigorous algorithmic thinking to mission-driven research challenges in data science and machine learning.

Education

Georgetown University

PHD, COMPUTER SCIENCE

- Advisor: Sasha Golovnev
- Research Interests: Algorithms, Data Structures, Discrete Math, Complexity
- GPA: 4.0

Washington D.C.
Fall 2021 - present

University of Rochester

HONORS BS, MATHEMATICS

- Computer Science Minor
- Thesis: Sampling Young Tableaux and Contingency Tables
- Take Five Scholar: The Role of Music in Video Games and Film
- Magna Cum Laude, GPA: 3.96

Rochester, NY
Fall 2016 - Spring 2021

Work Experience

Predict Health, Inc.

ML RESEARCHER

- Researched and developed retention models for Medicare populations to expand Predict Health's data science capabilities
- Built a custom Python package to ensure efficient, reusable capabilities for future retention projects
- Streamlined codebase to reduce project completion time from months to days
- Designed experiments to detect and mitigate data leakage in temporal prediction tasks
- Developed efficient pipeline for matching 100M records in Predict Health's data lake

Arlington, VA
May 2025 - Present

Georgetown University

COURSE DESIGNER AND INSTRUCTOR: *Intro to Programming in the Age of AI*

- Designed curriculum and materials for Georgetown's first AI-integrated Intro to Programming course
- Led twice-weekly lectures and workshops to teach students AI coding assistant usage and python fundamentals

Washington, D.C.
January 2026 - Present

Publications & Patents

Mark Bun, Mandar Juvekar, Samuel King. QMA Lower Bounds for Batch Verification via Approximate Degree. *Under review*.

Karthik Gajulapalli, Alexander Golovnev, Samuel King, and Sidhant Saraogi. Online Orthogonal Vectors Revisited. *Symposium on Discrete Algorithms (SODA)*, 2026.

Karthik Gajulapalli, Alexander Golovnev, and Samuel King. On the Power of Adaptivity for Function Inversion. *Conference on Information-Theoretic Cryptography (ITC)*, 2024.

Aniruddha Bapat, Andrew M. Childs, Alexey V. Gorshkov, Samuel King, Eddie Schoute, and Hrishee Shastri. Quantum routing with fast reversals. *Quantum*, 5, 533, 2021.

U.S. Patent Applications 18178491 & 18669111, filed Mar 3, 2023 & May 20, 2023.

Erin Bevilacqua, Samuel King, Jürgen Krietschgau, Michael Tait, Suzannah Tebon, and Michael Young. Rainbow numbers for $x_1 + x_2 = kx_3$ in $\mathbb{Z}_n$. *Integers: Electronic Journal of Combinatorial Number Theory*, 20(A50):A50, 2020.

Selected Coursework & Skills

Mathematics: Combinatorial Optimization, Spectral Graph Theory, Linear Algebra, Computational Statistics, Combinatorics, Algebra I & II

Computer Science: Algorithms, Data Structures, Sampling Algorithms, Analytical Methods, Cryptography, Parallel Algorithms, Databases, Gems of Theoretical CS

Artificial Intelligence: Experimental AI, Neural Networks, Artificial Intelligence, Philosophy of AI, AI Safety Fundamentals

Skills: Python (primary language), R, Mathematica, SQL (familiar languages), Git, $\text{\LaTeX}$, PyTorch, NetworkX, Pandas, scikit-learn, VSCode, Cursor, Databricks, Technical Communication

Projects

Reinforcement Learning for Ramsey Numbers Georgetown University
PERSONAL RESEARCH PROJECT Summer 2024

- Conducted research on RL approaches for finding counterexample graphs for lower bounds on Ramsey numbers
- Experimented with deep Q-learning, cross-entropy method, game-playing, and Monte Carlo tree search
- Experiments conducted using PyTorch, Gymnasium, Stable-Baselines3

Exploring Subitizing in Convolutional Neural Networks Georgetown University
NEURAL NETWORKS CLASS PROJECT Fall 2022

- Trained convolutional neural networks to count the number of simple shapes in an image
- Compared performance between classification and regression models
- Tested model robustness with different synthetic datasets

Stratego AI University of Rochester
HONORS SEMINAR RESEARCH PROJECT Spring 2018

- Implemented traditional state-space tree search algorithms to play the board game Stratego
- Experimented with using neural networks in Tensorflow to learn the state value function
- Experiments were unsuccessful, but the problem was solved by a team of 34 DeepMind scientists 4.5 years later

Teaching Experience

Intro to Programming in the Age of AI, Course Designer & Instructor Georgetown U, Spring '26
Gems of Theoretical CS, Teaching Assistant & Lecturer Georgetown U, Fall '25
CS Department Tutoring, Head Tutor & Program Founder Georgetown U, Spring & Fall '25
Data Structures, Guest Lecturer Amherst College, Fall '24
Introduction to Algorithms, Teaching Assistant & Lecturer Georgetown U, Fall '24
Math Methods for CS, Teaching Assistant & Lecturer Georgetown U, Spring '24
Intro to Databases, Teaching Assistant & Lecturer Georgetown U, Fall '23
New Horizons in Theoretical Computer Science, Teaching Assistant TTIC (Virtual), Summer '23
Graduate Algebra I, Teaching Assistant & Lecturer U of Rochester, Fall '20
Honors Calculus I & II, Teaching Assistant & Workshop Leader U of Rochester, Fall '17-Spring '20

Honors and Awards

- 2025 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department
- 2024 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department
- 2020 **Arthur S. Gale Memorial Prize**, University of Rochester Mathematics Department
- 2019 **Take Five Scholarship**, University of Rochester
- Deans' Award for Symposium Presentation**, University of Rochester
- Joint Mathematics Meetings 2019 Outstanding Poster Distinction**, JMM 2019